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TE A / B

EXPERIMENT NO 5

Aim: To study and Implement Infrastructure as a Service using AWS /Microsoft Azure.

Objective: To demonstrate the steps to deploy Web applications or Web services written in different languages on AWS Elastic Beanstalk/ Microsoft Azure App Service.

What is AWS Elastic Beanstalk?

Elastic Beanstalk is a platform within AWS that is used for deploying and scaling web applications. In simple terms this platform as a service (PaaS) takes your application code and deploys it while provisioning the supporting architecture and compute resources required for your code to run. Elastic Beanstalk also fully manages the patching and security updates for those provisioned resources.

There are many PaaS solutions in the cloud computing space including Redhat Open Shift, Google App Engine, Scalingo, Python Anywhere, Azure App Service, however AWS Elastic Beanstalk remains one of the leading PaaS choices among app developers.

There is no charge to use Elastic Beanstalk to deploy your applications, you are only charged for the resources that are created to support your application.

AWS Elastic Beanstalk allows you to quickly deploy applications and services without having to worry about configuring underlying resources, services, operating systems or web servers.

Elastic Beanstalk takes care of the hosting infrastructure, coding language interpreter, operating system, security, https service and application layer. All you need to worry about is writing your code. You can develop code in a number of languages which is then zipped up and the zip file is used when instantiating a new elastic beanstalk instance.

Supported language platforms include:

- Ruby
- Python
- PHP
- Packer Builder
- Glassfish
- Docker
- Tomcat

The web servers provisioned will be familiar to most web developers and include Apache, Tomcat, Nginx and IIS

You can still maintain control over the compute instance type used by elastic beanstalk when deploying your application to the cloud and you can also keep control over the database type and level of auto scaling required for your application.

You can access server log files of your deployed web application, update your application whenever required and enable HTTPS on the load balancer when required. Using the Elastic

Beanstalk platform delivers the opportunity to spend more time developing and less time managing your network, storage, o/s and compute runtimes as this is all handled by Elastic Beanstalk. This leads to

quicker deployment since all you need to do is package up your code, feed it to Elastic Beanstalk and the platform takes it from there.

You don't need to spend time selecting compute instances, database and storage requirements, security, monitoring services, load balancing resources and so on which leads to much faster deployment. You take care of the code and elastic beanstalk does the rest.

After deployment, the operations of your Elastic Beanstalk hosted applications are also easier. You no longer have to take on the role of monitoring servers, monitoring storage, managing network loads, keeping operating systems up to date since this is all taken care of by the platform.

Elastic Beanstalk Features

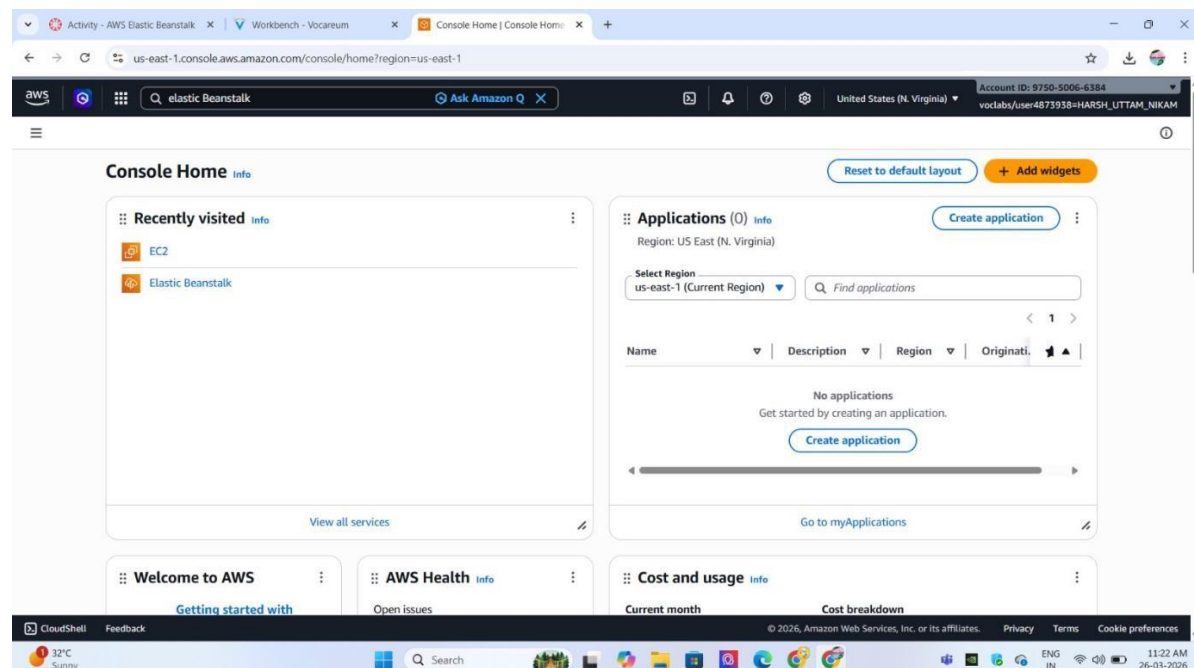
Elastic Beanstalk is possibly the simplest and fastest way to deploy web applications on AWS. It allows you to focus on writing code instead of provisioning and configuring AWS resources.

Elastic beanstalk handles the auto scaling of resources needed to support your deployed application as demand grows or shrinks.

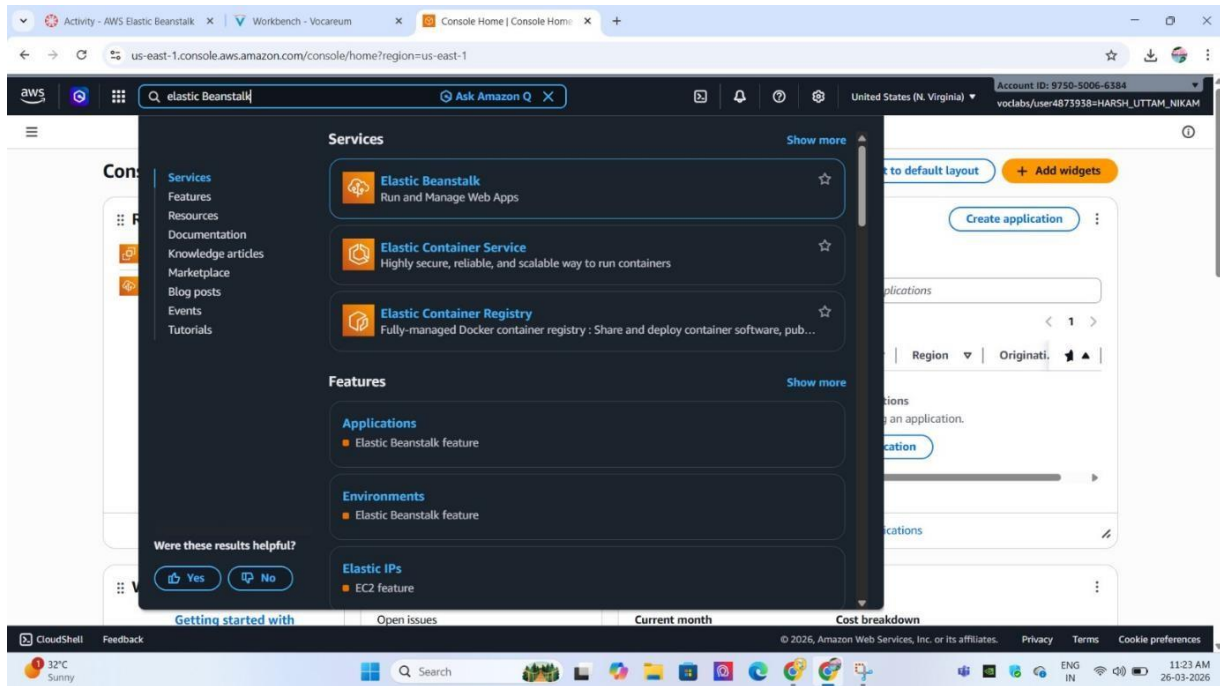
When Elastic Beanstalk analyzes your application and selects the resources that will be requires, it also allows you to step in and select alternative resources that may be better suited to anticipated use cases it may not know about. For example you could select a higher spec EC2 instance type that better suits your needs.

Steps:

Step 1: Open AWS Management Console

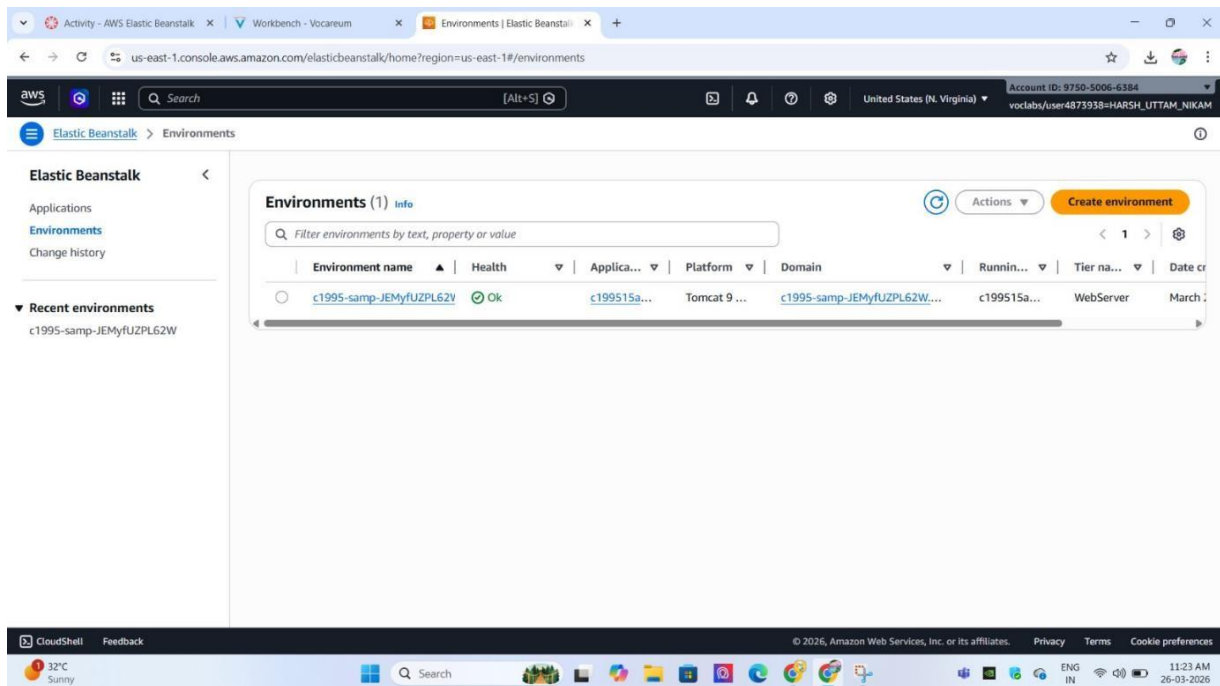


Step 2: Search for Elastic Beanstalk in search bar



Step 3: Click on Elastic Beanstalk service

Step 4: Open Environments section



Step 5: View created environment

Step 6: Click on Environment name

The screenshot shows the AWS Elastic Beanstalk console. The left sidebar contains navigation links for Applications, Environments, and Change history. The main content area displays the 'Environment overview' for 'c1995-samp-JEMyFUZPL62W'. The overview includes the following details:

- Health:** Ok
- Environment ID:** e-ussyqbeeuj
- Domain:** c1995-samp-JEMyFUZPL62W.eba-tkqcy5d.us-east-1.elasticbeanstalk.com
- Application name:** c199515a5102565114391340t1w975050066384-sampleApplication-M105NSCZVYBX
- Platform:** Tomcat 9 with Corretto 11 running on 64bit Amazon Linux 2023/5.12.0
- Running version:** c199515a5102565114391340t1w975050066384-sampleapplicationversion-i75fxhuo0ls-1
- Platform state:** Supported

Below the overview, there are tabs for Events, Deployments, Health, Logs, Monitoring, Alarms, Managed updates, and Tags. The 'Events' tab is selected, showing a list of events. The first event is dated March 26, 2026, 11:15:20 (UTC+5:30) and is an 'INFO' type, stating: 'Environment health has transitioned from info to Ok. Application update completed 58 seconds ago and took 73 seconds.'

Step 7: Check Environment overview (Health: OK)

Step 8: Click on Upload and deploy

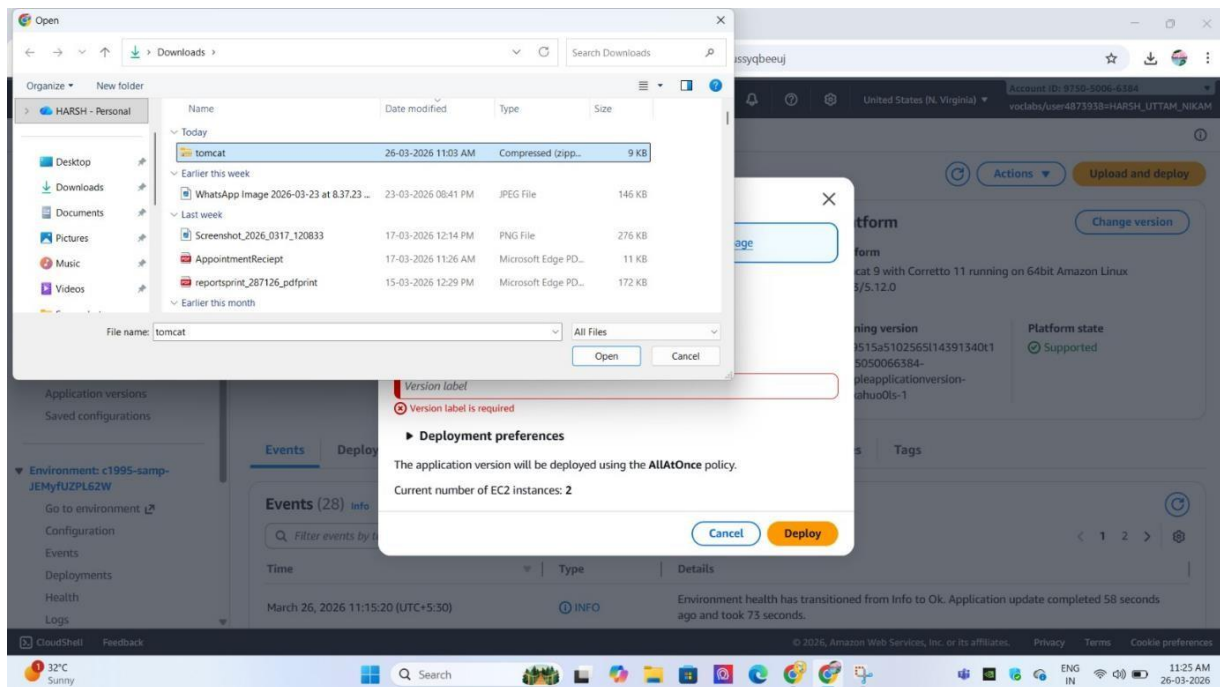
The screenshot shows the same AWS Elastic Beanstalk console environment overview. A modal dialog box titled 'Upload and deploy' is open in the center. The dialog contains the following information:

- A link: 'To deploy a previous version, go to the [Application versions page](#)'
- Upload application:** A 'Choose file' button. Below it, a note states: 'File must be less than 500MB max file size'.
- Version label:** A text input field with the placeholder 'Version label'. A note below it says: 'Unique name for this version of your application code.'
- Deployment preferences:** A section stating: 'The application version will be deployed using the **AllAtOnce** policy.' and 'Current number of EC2 instances: 2'.

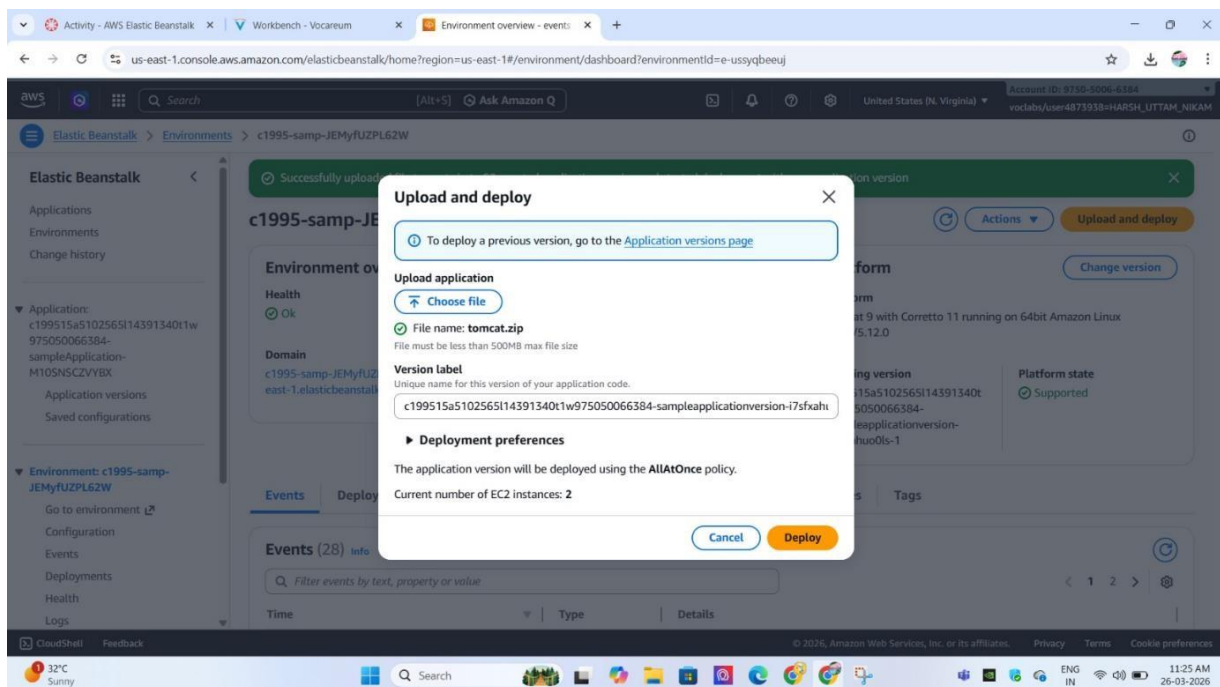
At the bottom of the dialog are 'Cancel' and 'Deploy' buttons.

Step 9: Click on Choose file

Step 10: Select your .zip file (tomcat.zip)

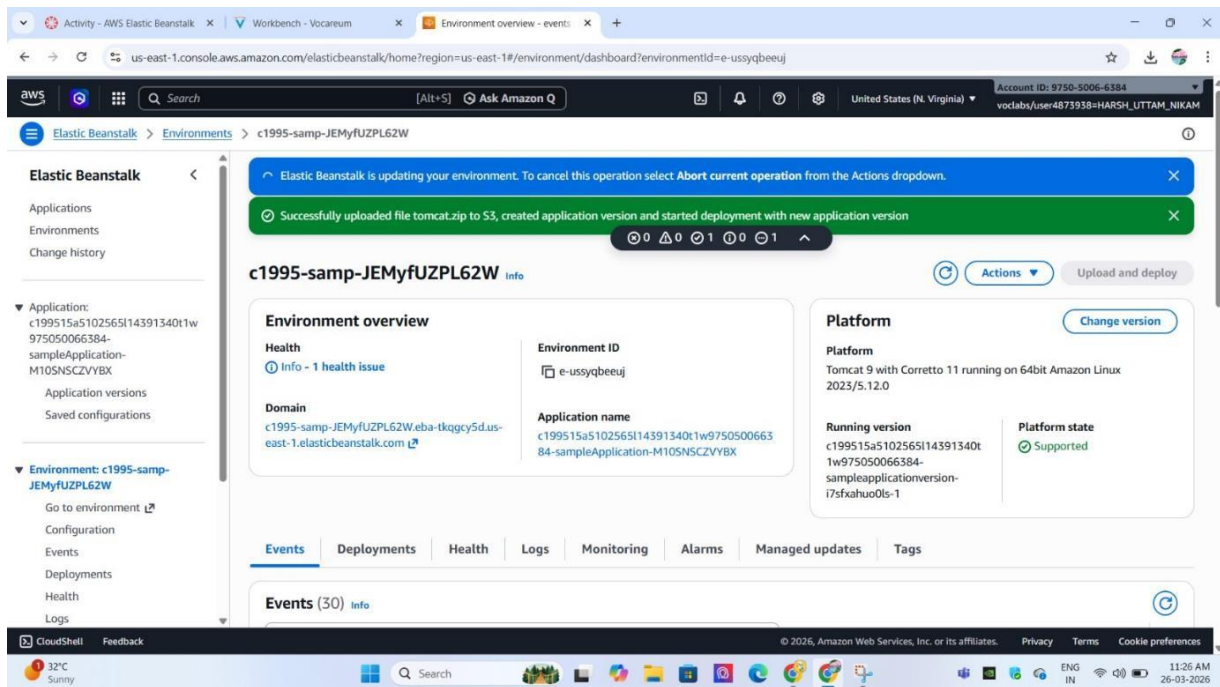


Step 11: Enter Version label

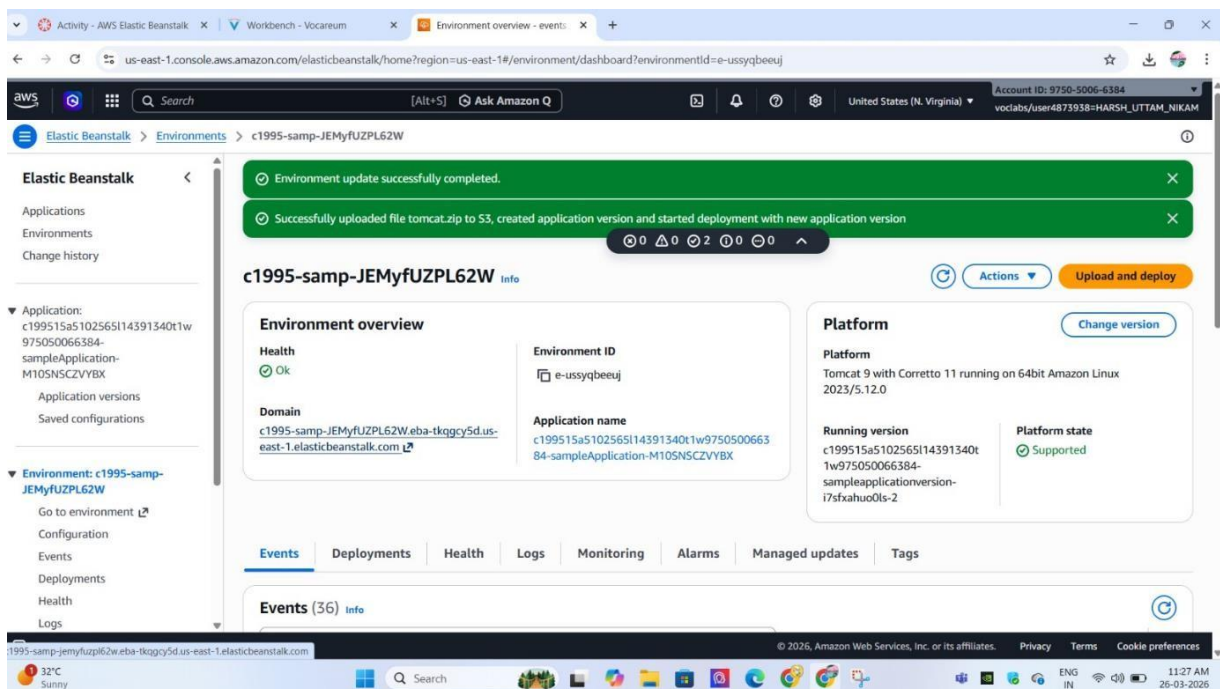


Step 12: Click on Deploy

Step 13: Wait for deployment process



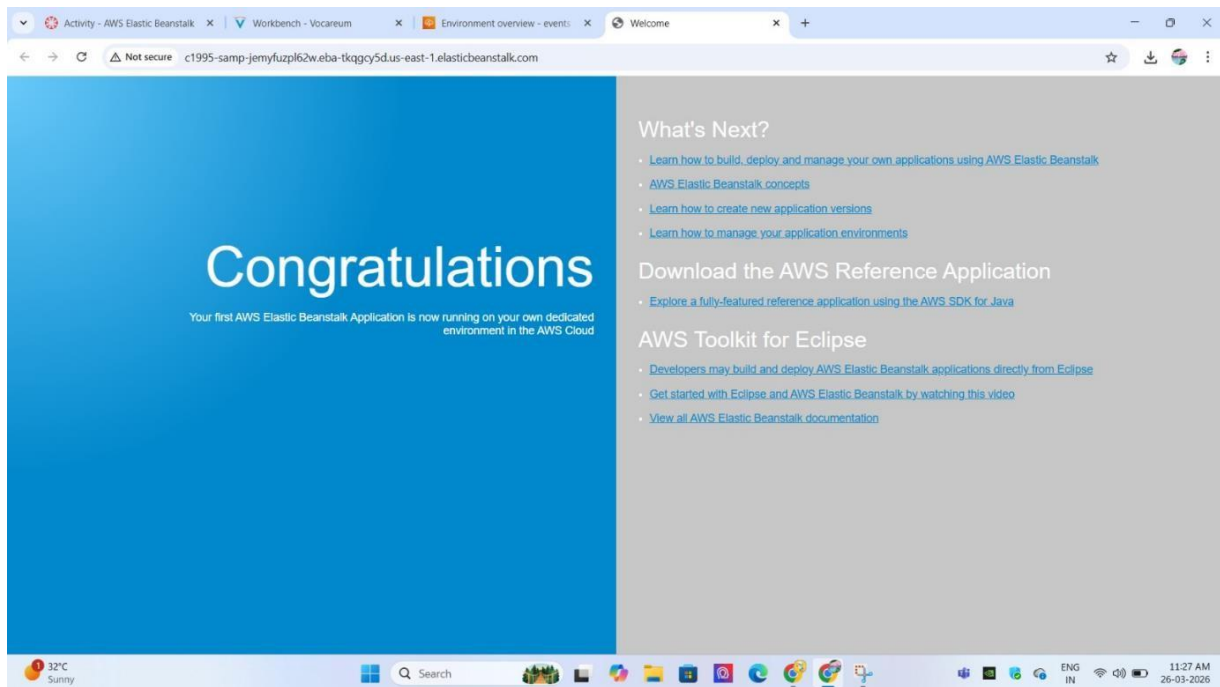
Step 14: Check Success message (uploaded & deployed)



Step 15: Wait until Environment health becomes OK

Step 16: Click on Application URL (Domain link)

Step 17: Open deployed web application



Step 18: Verify Application Congratulations page displayed

Conclusion :

Thus, we have successfully studied and implemented Platform as a Service (PaaS) using AWS Elastic Beanstalk.